

Arun Kumar Kumari Arumugam

AI Trainer | LLM Evaluation | Medical SFT & NLP Annotation | Agentic HITL

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OpenTrain: app.opentrain.ai/labeler-profile/arun-k-35 | GitHub: github.com/arunkumar4189/AI-ML-Projects

PROFILE

Distinguished Engineer at Verizon (15+ years building distributed platforms) and M.Tech AI/ML candidate at BITS Pilani (2024AC05045). I create instruction data, score LLM and agent outputs against rubrics, annotate medical and dialogue text, and run human-in-the-loop gates on production multi-agent systems. Best fit: LLM evaluation, SFT/preference data, medical or telco NLP, code-aware review, and agent-trace QA -- not high-volume bounding-box piecework.

AI TRAINING AND ANNOTATION

Verizon -- GenAI / agentic HITL and LLM evaluation

Chennai | 2024 -- Present | Internal tooling | text, code, documents | eval, HITL, prompts, safety

- HITL validation for Agentic Universe (React, FastAPI, MCP, PostgreSQL): workflow review, tool-call checks, approval gates.
- Scored LangGraph SRE triage traces that cite OpenSearch logs, git diffs, Jira issues, and Kubernetes change windows.
- Evaluated MCP Agent Companion outputs for planning, tests, and code review across Java, Python, and TypeScript.
- Rubrics: factuality, groundedness/citations, instruction-following, and unsafe or unapproved actions.

BITS Pilani M.Tech -- LLM / Generative AI (Group 101)

2025 -- 2026 | Hugging Face, PEFT QLoRA, JSONL | medical documents | SFT, fine-tuning, prompt eval

- Built medical instruction-response pairs from a PDF corpus; QLoRA-tuned GPT-2 Medium (4-bit NF4, r=16, alpha=32).
- Prompting system with a shared eval harness: zero/few-shot, self-critique, JSON function-calling, CoT/ToT/self-consistency/ReAct, and a strategy router.
- Scored answers for domain fidelity, JSON schema validity, token/latency cost, and critique quality.

BITS Pilani M.Tech -- NLP, IR, conversational AI

2025 -- 2026 | NLTK, scikit-learn, Hugging Face | text, documents | NER, classification, QA, dialogue

- Medical QA over five clinical sources (diabetes, hypertension, COVID-19, asthma, appendicitis) with retrieval, Flan-T5, BART intent, and a verify-and-correct loop.
- University-helpdesk gold set: 12 intents, 80+ utterances, slot/NER gazetteers, multi-turn state, safety escalation.
- POS + NER on Airbnb reviews; restaurant sentiment with syntactic negation; news taxonomy (Naive Bayes/Rocchio); clinical IR and spelling correction.

BITS Pilani M.Tech -- CV, classical ML, RL, LLMops

2025 -- 2026 | CNN/MLP, MLflow, Gradio, DistilBERT+LoRA | image, text, tabular

- Image-level classification (cats vs dogs); Breast Cancer Wisconsin diagnostics; Apache Ant software-defect labels.
- SupportSense: Banking77 intent plus sentiment, summary, FAQ QA, image caption/defect, draft reply; telco churn (7,043 rows) reviewed in MLflow/Streamlit.
- Bandit and Q-learning policy comparison for preference/reward analysis (coursework, not a vendor RLHF contract).

PROFESSIONAL BACKGROUND (DOMAIN CONTEXT)

Verizon -- Distinguished Engineer, Software Development

Chennai | Sep 2025 -- Present

GenAI and agentic roadmap; launched Agentic Universe; architecture board for B2B View Together; org-wide Claude Code / Copilot and MCP agent suite.

Verizon -- Principal Engineer / Solutions Architect

Jan 2021 -- Sep 2025

B2B SMB & NSE ordering (Spring Reactive, GraphQL, Cassandra); SMB Enrollment and Add-A-Line (OCR ScanID, PEGA); bulk engine 10,000+ orders per payload; 99.99% uptime with New Relic.

Earlier roles

Verizon Lead/Architect 2016--2021; Tech Mahindra 2014--2016; Wipro 2010--2013

Event-driven REST, Quote-to-Order automation, Oracle ATG commerce, WebLogic, catalog and order repositories.

MATCHING SKILLS

Data: text, documents, code, images, medical. Tasks: LLM eval/rating, SFT, LoRA/QLoRA, NER, intent/slots, QA/RAG groundedness, sentiment, classification, prompt/red-team, HITL agent traces, image-level labels. Tools: Hugging Face, PEFT, LangGraph, MCP, FastAPI, MLflow, Gradio, Streamlit, NLTK, scikit-learn, OpenSearch, Jupyter. Code: Python, Java, TypeScript, JavaScript, SQL. Human language: English (fluent).

EDUCATION

- M.Tech, Artificial Intelligence & Machine Learning -- BITS Pilani -- 2025-2027 (expected) -- ID 2024AC05045
- B.E., Computer Science & Engineering -- Anna University, Chennai -- 2010 -- 77.06%